

KAREEM SALAH

Mechatronics Engineer

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Experience

Birr Machines

Design Automation Engineer – SOLIDWORKS API / C#

04/2026 – Present

- Co-initiated and currently developing a SOLIDWORKS API-based automation system in C# to auto-generate motor designs (parts, assemblies, and drawings)
- Integrated PDM system for version control and automated deliverable generation including PDFs and BOMs
- Reduced manual design iteration time by scripting parametric model generation across the full design-to-documentation pipeline

Mechanical Design Engineer – Rotating Machines

01/2025 – 04/2026

- Designed custom motors and generators ranging from 20 kW to 30 MW using SOLIDWORKS
- Created and updated 3D CAD models, assemblies, and detailed manufacturing drawings
- Collaborated with electrical and manufacturing teams to align mechanical design with system requirements

Takamul Education

Robotics R&D Engineer

09/2025 – 12/2025

- Developed modular robotics control systems with voice command, OTA firmware updates, and multi-device interfaces
- Designed and prototyped educational robotic platforms with real-time sensor interaction and feedback
- Delivered robotics training sessions and workshops for students and instructors on developed hardware/software systems
- Conducted reverse engineering on existing kits to improve reliability and performance of deployed platforms

NISBAU EGYPT

Mechanical Design & manufacturing Engineer

01/2025 - 09/2025

- Design and redesign German concrete plants structure for manufacture and assembly based on location and needs
- Conduct research and development of new plants to enhance efficiency and sustainability.

Ever Egypt

EVER Egypt dynamic track (urban electric vehicle)

04/2022 - 09/2025

- joined the competition twice making a total of two fully one seat electric urban vehicle , once as a suspension head and the other as a mechanical advisor helping all mechanical systems
- As the suspension system co-head, collaborated with the team to design, manufacture, and test the suspension system for a full vehicle
- Utilized SOLIDWORKS for design, assembly & simulation, optimizing kinematics for kinematics simulation
- Served as a mechanical student advisor for all systems integration, kinematics and dynamics

Shell eco marathon

Shell Eco-Marathon Racing team leader

01/2023 - 05/2023

- Led a multidisciplinary team in designing, building, and optimizing an ultra-efficient electric vehicle for the Shell Eco-Marathon, a globally recognized competition focused on energy efficiency and innovation.
- Directed vehicle system optimizations, including aerodynamics, weight reduction, and powertrain efficiency, achieving compliance with stringent competition standards.

ASME

ASME HPVC (Human Powered Vehicle Competition)

01/2022 - 05/2022

- Led design, manufacturing, and testing of suspension system for a hybrid human/electric powered vehicle, implementing innovative double wishbones and direct damper mechanisms improving handling and performance.

Grenium Startup

Genium co-founder: Hydrogen Generator System

02/2022 - 05/2025

- Co-founded Genium, a startup pioneering a sustainable HHO electrolyzer, securing 1st place in entrepreneurship competitions for its innovative hydrogen generation solution.
- Designed the mechanical structure using SOLIDWORKS, featuring corrosion-resistant stainless steel electrodes, optimized cell stack assembly, and a durable enclosure for safety and efficiency
- Designed the mechanical enclosure and integrated electrical components, with pressure, current, and temperature and other sensors with precise prototyping

ASME

ASME Assuit student section president

05/2023 - 05/2024

- Leadership and Coordination: Managing the team, of over 80+ members, organizing events, and ensuring smooth collaboration among
- Technical Development: Overseeing technical workshops, projects, and competitions to enhance members' skills.
- competitions : managed to get the team to participate in multiple robotics&vehciles big competitions(IRC sumo robot-puck collector robot ,ASME XRC autonomus vehicle -lunarlander-ocean cleaner ,shell eco marathon and more) as a manager and a technical member earning multiple international and national rankings and achievements

Education

Mechatronics department

Assuit University , faculty of engineering

07/2019 - 10/2024

- Mechanical & Mechatronics

Language school

Modern Manar language school

01/2005 - 09/2019

Find me online

Linkedin

[In/kareem-salah-933435221/](https://www.linkedin.com/in/kareem-salah-933435221/)

Github

www.github.com/kareem0o0

Training / Certificates

SOLIDWORKS CHAMPION

CSWA / CSWP : Certified SOLIDWORKS professional

HA Consulting Group: Automation ,PLC and classic control

Al-Mansour Automotive Internship: One month summer internship at Al-Mansour Automotive Chevrolet service department

Languages

Arabic Native ●●●●●

English Proficient ●●●●○

German Intermediate ●●○○○

Skills

Solidworks • Autodesk Inventor • DFM & DFA • Vehicle design • Product design • Control • Simulation • Embedded systems • Leadership

Key Achievements

ASME XRC International

Ranked 1st internationally – Autonomous Ocean Cleanup Robot; Ranked 2nd internationally – Autonomous Lunar Lander

EVER Egypt Autonomous Track

Ranked 2nd in Egypt

EVER Egypt Dynamic Track

Ranked 5th in Egypt

Robotics Challenge Egypt

Ranked 1st – Puck Collector Robot; Ranked 2nd – Sumo Robot

ASME Student Sections

ASME Membership Growth Award – Middle East & Africa (as chairman)